

Question	Marking Guidance	Mark	Comments
08.1	1. (After 11 years) there is a large decrease in the number of helper T-cells OR (After 11 years) there are few/low number of helper T-cells; 2. (So) fewer B cells divide by mitosis OR (So) fewer B cells undergo clonal selection; 3. (So) fewer plasma cells produce antibodies;	3 (3 x AO3)	1. Allow other words for large eg significant/big 2.and 3. Accept 'no' for fewer 2. Accept clonal expansion

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08.2	Mark as pairs 1 and 2 or 3 and 4 or 5 and 6 or 7 and 8. Justification must match action. 1. Lift lid of (agar) plate at an angle; 2. To prevent contaminating the dish OR Prevent spread of bacteria outside the dish; 3. Flame the spreader OR Use a sterile spreader OR Flame the neck of the container of the culture; 4. To maintain a pure culture OR To prevent contaminating the dish OR Prevent spread of bacteria outside the dish; 5. Work close to a Bunsen burner/flame; 6. To create a convection current to carry other contaminants away/prevent them landing on the dish OR To create upward air movement to carry other contaminants away/prevent them landing on the dish; 7. Place used pipette/spreader into (container of) disinfectant; 8. To ensure no transfer of bacteria to bench;	2 max (2 x AO3)	1. Accept lift lid slightly OR keep lid over plate OR open lid of petri dish as little as possible 3. Accept loop for spreader Ignore 'wash hands'